

Find factor pairs systematically

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: List every factor pair of 72. Copy or complete the first step.

2. List all factors of 48.

3. How many factors does 36 have?

4. Miro says: Miro stops checking factor pairs after reaching 5. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: List every factor pair of 72. Copy or complete the first step.
Answer 1 and 72; 2 and 36; 3 and 24; 4 and 18; 6 and 12; 8 and 9.
2. List all factors of 48.
Answer 1, 2, 3, 4, 6, 8, 12, 16, 24, 48.
3. How many factors does 36 have?
Answer 9 factors.
4. Miro says: Miro stops checking factor pairs after reaching 5. Is this correct? Explain.
Answer Continue until the first factor meets or passes its paired factor.

Find factor pairs systematically

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. List every factor pair of 72.

6. Check this result using an inverse, estimate or known fact: How many factors does 36 have?

2. List all factors of 48.

7. Find a number below 100 with exactly 10 factors.

3. How many factors does 36 have?

8. Identify and correct this misconception: Miro stops checking factor pairs after reaching 5.

4. Solve this independently and show every stage: List every factor pair of 72.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: List all factors of 48.

10. Write and solve a similar question to: How many factors does 36 have?

Teacher answers and checking prompts

1. List every factor pair of 72.
Answer 1 and 72; 2 and 36; 3 and 24; 4 and 18; 6 and 12; 8 and 9.
2. List all factors of 48.
Answer 1, 2, 3, 4, 6, 8, 12, 16, 24, 48.
3. How many factors does 36 have?
Answer 9 factors.
4. Solve this independently and show every stage: List every factor pair of 72.
Answer 1 and 72; 2 and 36; 3 and 24; 4 and 18; 6 and 12; 8 and 9.
5. Use a second method or representation for: List all factors of 48.
Answer Expected result: 1, 2, 3, 4, 6, 8, 12, 16, 24, 48. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: How many factors does 36 have?
Answer Expected result: 9 factors. A valid check must be shown.
7. Find a number below 100 with exactly 10 factors.
Answer 48 is one example.
8. Identify and correct this misconception: Miro stops checking factor pairs after reaching 5.
Answer Continue until the first factor meets or passes its paired factor.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: How many factors does 36 have?
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Find factor pairs systematically

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Find a number below 100 with exactly 10 factors.

6. Create a new example that tests the same idea as: How many factors does 36 have?

2. Prove the answer to this question, rather than only calculating it: List every factor pair of 72.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: List all factors of 48.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 9 factors.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro stops checking factor pairs after reaching 5.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Find a number below 100 with exactly 10 factors.
Answer 48 is one example.
2. Prove the answer to this question, rather than only calculating it: List every factor pair of 72.
Answer Expected result: 1 and 72; 2 and 36; 3 and 24; 4 and 18; 6 and 12; 8 and 9. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: List all factors of 48.
Answer Expected result: 1, 2, 3, 4, 6, 8, 12, 16, 24, 48. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 9 factors.
Answer One valid reconstruction is: How many factors does 36 have? Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro stops checking factor pairs after reaching 5.
Answer Continue until the first factor meets or passes its paired factor.
6. Create a new example that tests the same idea as: How many factors does 36 have?
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.